YEAR 11 PHYSICS, AQA GCSE, LESSON 10 OF 10

Atoms & Nuclear Radiation.

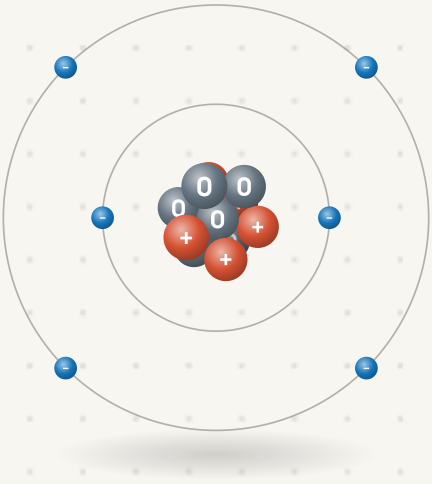
Lesson 10 Contamination and irradiation

Supermarkets can sell food treated with gamma rays to kill bacteria. Does eating it make you radioactive?

IN THIS LESSON YOU WILL

- explain the difference between irradiation and contamination
- compare the hazards of alpha, beta and gamma inside and outside the body
- describe precautions for working with radioactive sources
- explain why studies of radiation effects are published and peer reviewed

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

How this booklet works.

This is the booklet for **Lesson 10** of the Atoms and Nuclear Radiation unit. It builds on the lessons before it; the "Do now" questions and the quick answers at the back help you check what you remember. Keep it: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
Explanations, diagrams and worked examples. Read this again when you revise.

TRY IT
Practice that builds up: first with support, then on your own, then a challenge.

LAB
Record what you find in a simulation or a model experiment.

EXAM CORNER
AQA-style questions with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an "I can" check to colour in.

! **Revising at home?** (1) Read the Learn it pages. (2) Cover them and answer the "Cover the page" questions in Revise it. (3) Check with the quick answers at the back. (4) Go through the matching slides in the teaching deck, and try the simulations.

The route through the unit (this lesson is highlighted)

01 Inside the atom

02 Isotopes and ions

03 From plum pudding to the nucleus

04 Energy levels and the nucleus

05 Radioactive decay

06 Nuclear equations

07 Properties and uses of radiation

08 Half-life and random decay

09 Half-life calculations

10 **Contamination and irradiation**
Two ways to be exposed, hazards inside and outside the body, staying safe, peer review (slides 38–41)

Teaching deck: **panphy.app/gcsephy/decks/atoms-and-radiation**. Simulations: **panphy.app/simulations/atomic_models.html** and **panphy.app/simulations/nuclear_decay.html**

LESSON

10 Contamination and irradiation.

AQA Physics 4.4.2.4; Combined Science 6.4.2.4

BIG QUESTION

Supermarkets can sell food treated with gamma rays to kill bacteria. Does eating it make you radioactive?

BY THE END YOU CAN

- explain the difference between irradiation and contamination
- compare the hazards of alpha, beta and gamma inside and outside the body
- describe precautions for working with radioactive sources
- explain why studies of radiation effects are published and peer reviewed

irradiation | contamination | hazard | precaution | dosimeter | peer review

SLIDES AND SIMULATIONS

Teaching deck, slides 38–41

Contamination and irradiation, the quick check and the key words.

panphy.app/gcsephy/decks/atoms-and-radiation/#38

RETRIEVE

Do now

From earlier lessons. No notes!

1 A source has a half-life of 3 hours. What fraction is left after 9 hours?

⅛ (3 half-lives).

2 Which radiation is the most penetrating?

Gamma.

3 What stops beta radiation?

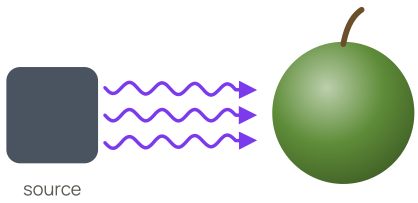
A few mm of aluminium.

EXPLAIN

Two different things

Irradiation and contamination

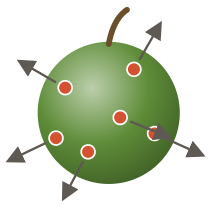
IRRADIATION



exposed to radiation

does NOT become radioactive

CONTAMINATION



radioactive atoms on or in it

keeps emitting until removed

	Irradiation	Contamination
What is it?	exposing an object to nuclear radiation from outside	unwanted radioactive atoms **on or in** an object
Does the object become radioactive?	**no**	it contains radioactive atoms, so it gives out radiation
When does the exposure stop?	when the source is removed or shielded	only when the radioactive atoms are removed (or have decayed)
Example	gamma rays used to sterilise food or medical equipment	breathing in radon gas; radioactive dust on skin

! An irradiated object does **not** become radioactive. A contaminated object keeps giving out radiation until the radioactive atoms are removed.

How big is the hazard?

Ionising radiation can damage cells and the DNA inside them: cells may die or start to divide out of control (cancer). The hazard depends on where the source is and which radiation it gives out.

Radiation	Source outside the body	Source inside the body
Alpha	low hazard: stopped by the outer layer of skin	**most hazardous**: strongly ionising, all its energy goes into nearby cells
Beta	can pass through skin to tissues below	hazardous: moderately ionising
Gamma	**most hazardous** outside: reaches organs inside	least hazardous inside: much of it passes out of the body

Staying safe

PRECAUTIONS WITH RADIOACTIVE SOURCES

- Handle sources with **tongs** or wear **gloves**: keeps the source away from the skin and stops contamination.
- Keep your **distance** and point sources away from people.
- Keep **exposure time** as short as possible.
- Store sources in **lead-lined boxes**, which absorb the radiation.
- Workers wear a **dosimeter badge** to measure how much radiation they receive.

Science you can trust

In the 1920s, workers painting watch dials with glowing radium paint licked their brushes to make a fine point. Many became seriously ill. Published studies of their health helped show how dangerous radioactive contamination is, and safety rules changed. It is important that findings about the effects of radiation on people are **published** and shared, so other scientists can check them by **peer review**.

PRACTISE

Try it

1 Is each example **irradiation** or **contamination**? Circle one for each.

A
Surgical instruments are sterilised with gamma rays.
irradiation
contamination

B
A person breathes in radon gas.
irradiation
contamination

C
Radioactive dust sticks to a worker's gloves.
irradiation
contamination

D
Strawberries are treated with gamma rays to keep them fresh.
irradiation
contamination

2 Explain why Leo is wrong.

Leo: Food treated with gamma rays becomes radioactive, so it is dangerous to eat.

This is irradiation: the food is exposed to radiation but no radioactive atoms are added, so it does not become radioactive and is safe to eat.

3 SUPPORTED Explain how each precaution reduces the risk.

Precaution	How it reduces the risk
Using tongs	keeps the source away from the hands; less radiation reaches them and no contamination
Storing the source in a lead-lined box	lead absorbs the radiation
Keeping exposure time short	less radiation is received in total
Wearing a dosimeter badge	measures the dose received, so it can be kept below safe limits

4 Explain why an alpha source is the least hazardous outside the body but the most hazardous inside it.

Outside: alpha is stopped by the outer layer of skin (or a few cm of air), so it cannot reach living cells. Inside: nothing shields the cells, and alpha is strongly ionising, so it transfers all its energy to nearby cells and causes a lot of damage.

5 Why is it important that studies of the effects of radiation on people are published? Tick **one** box.

- [x] So that other scientists can check the findings (peer review).
- [ ] So that the scientists become famous.
- [ ] So that the results can never be changed.
- [ ] So that radiation becomes safe.

6 CHALLENGE Radon-222 is a gas given off by some rocks, and it emits alpha particles. Explain why radon in homes is a health risk, even though alpha cannot pass through skin.

Radon gas is breathed in, so it contaminates the lungs: the source is inside the body. The alpha particles are strongly ionising and damage lung cells directly, which can cause cancer.

AQA-STYLE

Exam corner

Write in full sentences. Show every step of a calculation.

Get the marks: Define both words, and make the difference clear: outside source vs radioactive atoms on or in the object.

7 Explain the difference between irradiation and contamination. [2 marks]

Irradiation: an object is exposed to radiation from a source outside it and does not become radioactive (1). Contamination: radioactive atoms get onto or into an object, so it keeps emitting radiation (1).

8 A technician uses a sealed gamma source. Describe two precautions the technician should take, and explain how each one reduces the risk. [4 marks]

Any two, each with a reason: use tongs / keep a distance (1) so less radiation reaches the body (1); keep exposure time short (1) so the total dose is lower (1); store in a lead-lined box (1) because lead absorbs gamma (1); wear a dosimeter (1) to monitor the dose (1).

REVIEW LATER

Revise it

Use this box after the lesson and before tests.

SUMMARY

- Irradiation: exposed to radiation; does not become radioactive.
- Contamination: radioactive atoms on or in an object; it keeps emitting until they are removed.
- Outside the body, gamma (and beta) are most hazardous; inside, alpha is most hazardous.
- Precautions: tongs, distance, short time, lead shielding, dosimeter.
- Findings about radiation's effects are published and peer reviewed.

COVER THE PAGE AND ANSWER

1 Does irradiated food become radioactive?

2 What is contamination?

3 Which radiation is most hazardous inside the body?

4 Why are results about radiation's effects published?

I CAN…

- explain the difference between irradiation and contamination
- compare the hazards inside and outside the body
- describe precautions and how each reduces the risk
- explain why findings are published and peer reviewed

REFERENCE

Toolkit.

Learn these: exam questions expect you to recall them without a data sheet.

The three particles

Lesson 1

Particle	Charge	Mass	Where
Proton	+1	1	nucleus
Neutron	0	1	nucleus
Electron	−1	very small	energy levels

Reading the symbol

Lesson 1

$^{A}_{Z}X$ e.g. $^{23}_{11}Na$

- Z, atomic number = protons
- A, mass number = protons + neutrons
- neutrons = A − Z; in an atom, electrons = Z
- Atom radius ≈ 1×10^{-10} m; nucleus radius < 1/10 000 of that

Isotopes and ions

Lesson 2

- **Isotopes**: same number of protons, different numbers of neutrons.
- **Positive ion**: an atom that has lost one or more outer electrons. The nucleus is unchanged.

Alpha scattering

Lesson 3

Observation	Conclusion
most straight through	mostly empty space
some deflected	positive centre
a very few bounced back	mass in a tiny centre

Models and energy levels

Lesson 4

- 1897 electron → 1904 plum pudding → 1911 nuclear → 1913 Bohr → 1932 neutron (Chadwick)
- **Absorb** EM radiation → higher level, further out.
- **Emit** EM radiation → lower level, closer in.

Nuclear radiation

Lesson 5

- Unstable nuclei decay at **random** to become more stable.
- **Activity**: decays per second, in becquerel (Bq). **Count rate**: decays recorded per second by a detector.
- α: 2p + 2n; β: fast electron, a neutron → proton; γ: EM radiation; n: a neutron

Nuclear equations

Lesson 6

Emitted	Symbol	Mass no.	Atomic no.
alpha	$^{4}_{2}\mathrm{He}$	− 4	− 2
beta	$^{0}_{-1}\mathrm{e}$	no change	+ 1
gamma	γ	no change	no change
neutron	$^{1}_{0}\mathrm{n}$	− 1	no change

Top numbers balance; bottom numbers balance.

$$^{226}_{88}\mathrm{Ra} \rightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\mathrm{He}$$

$$^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N} + {}^{0}_{-1}\mathrm{e}$$

Properties of radiation

Lesson 7

	Alpha	Beta	Gamma
Ionising	strongly	moderately	weakly
Range in air	a few cm	about 1 m	very far
Stopped by	paper, skin	a few mm of aluminium	reduced by thick lead or concrete
Typical use	smoke alarm	thickness gauge	medical tracer

More ionising → less penetrating.

Half-life from a graph

Lesson 8

1. Read the start value.
2. Halve it.
3. Across to the curve, down to the time axis.
4. Check with a second halving.

Half-life calculations

Lesson 9

- Number of half-lives = time ÷ half-life.
- After n half-lives, $(½)^n$ is left: ½, ¼, ⅛, 1/16…
- Net decline = initial − final; as a ratio of the initial, $1 - (½)^n$: ½, ¾, ⅞…

Contamination and irradiation

Lesson 10

- **Irradiation**: exposed to radiation; does not become radioactive.
- **Contamination**: radioactive atoms on or in an object.
- Outside the body γ is most hazardous; inside, α is.
- Tongs, distance, short time, lead, dosimeter.

REFERENCE

Key words from this lesson.

Learn the meaning and spelling. In exams, precise words earn marks: say “nucleus”, not “middle”, and “unstable nucleus”, not “unstable atom”.

Contamination: Unwanted radioactive atoms on or in an object.

Dosimeter: A badge that measures how much radiation a worker receives.

Irradiation: Exposing an object to nuclear radiation. The object does not become radioactive.

CHECK YOURSELF

Quick answers.

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

DO NOW

1. ⅛ (3 half-lives).
2. Gamma.
3. A few mm of aluminium.

COVER THE PAGE

1. No.
2. Unwanted radioactive atoms on or in an object.
3. Alpha.
4. So other scientists can check them (peer review).